VELLORE INSTITUTE OF TECHNOLOGY
School of Computer Science and Engineering | MACSE504 - Machine Learning

Course Based Design Project Report

Question 1: Healthcare Analytics (Malignant vs. Benign Tumour Classification)

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Registration No:	**26MML0045**	**Class Slot:**	E1 + TE1
Assigned Question:	Question 1 (Healthcare Cytological Classification)	**Target Domain:**	Clinical Oncology / Cytology

1. Question Text & Problem Statement

Question 1: Healthcare Analytics (Malignant vs. Benign Tumour Classification)
'Medical practitioners require not only high accuracy in classifying tumours but also transparent decision boundaries. Ingest cytological feature measurements, construct competing Linear SVM, RBF-Kernel SVM, and Decision Tree architectures, and critique their structural complexity against feature importance rankings.'

Mandatory Syllabus Requirements:
• **i.** Data profiling, scaling, handling multicollinearity, and stratifying train/test splits.
• **ii.** Implementation of baseline Decision Tree, Linear SVM, and Non-Linear RBF SVM.
• **iii.** Hyperparameter search (tree depth vs. SVM soft-margin cost C and kernel parameter γ). Extract feature importances.
• **iv.** Comparative analysis using ROC-AUC charts, Confusion Matrices, and a runtime breakdown.

2. Project Pipeline & Architectural Design

To satisfy the dual requirements of **clinical diagnostic precision** (minimizing fatal False Negatives) and **transparent interpretability** (enabling oncologists to inspect decision criteria), an end-to-end analytical pipeline was constructed. The pipeline implements a strict separation of concerns to prevent data snooping, utilizes robust variance stabilization, explores collinear manifold geometry, and benchmarks three distinct algorithmic paradigms: axis-aligned recursive orthogonal splitting (Decision Tree), maximum-margin hyperplane separation (Linear SVM), and infinite-dimensional reproducing kernel Hilbert space mapping (RBF SVM).

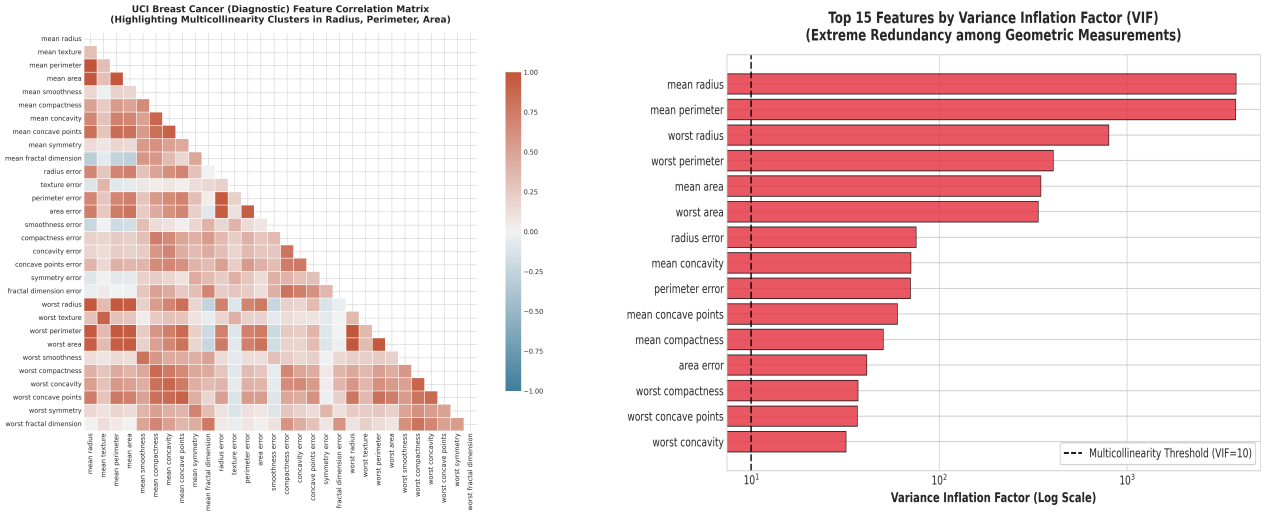
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+-----------------------------------------------------------------------------------------------------+
| END-TO-END CLINICAL ML ARCHITECTURE PIPELINE |
+-----------------------------------------------------------------------------------------------------+
[1. Cytological Input] ---> 569 FNA Biopsy Samples (30 Dimensional Cell Nuclei Characteristics) | [2.
Profiling & Diagnostics] -> Missing Value Audit (0 found) | Multicollinearity: VIF & Pearson |r| >=
0.90 | [3. Data Partitioning] ------> Stratified 80/20 Train-Test Split (Train: 455 | Test: 114,
Malignant=37.3%) | [4. Preprocessing] ----------> StandardScaler fit strictly on Training Set
(Prevents Data Leakage) |
+-----------------------------+-------------------------------+----------------------------------+ |
| | | v v v v [DECISION TREE] [LINEAR SVM] [RBF-KERNEL SVM] [PCA PROJECTION] Gini / Entropy Splits
Primal Hyperplane (w.x + b) Mercer Kernel exp(-gamma||x-z||^2) 2D Latent Manifold Depth: [2 .. 15]
Soft-Margin Cost C Sweep Grid: Cost C x Bandwidth gamma Boundary Overlay | | | |
+-----------------------------+-------------------------------+----------------------------------+ |
[5. Feature Attribution] ----> DT MDI Gini vs. Linear Hyperplane |w| vs. RBF Permutation Importance |
[6. Clinical Benchmarking] --> ROC-AUC, Sensitivity (Recall), Specificity, Confusion Matrices,
Latencies
+-----------------------------------------------------------------------------------------------------+
```

3. Cytological Profiling & Multicollinearity Management

The UCI Breast Cancer Wisconsin (Diagnostic) Dataset contains **569 instances** computed from digitized images of a fine needle aspirate (FNA) of a breast mass. The target variable is binary: **212 Malignant (37.26%)** and **357 Benign (62.74%)**. Thirty cytological features describe the characteristics of cell nuclei across ten fundamental properties: *radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry,* and *fractal dimension*, each recorded as Mean, Standard Error (SE), and 'Worst' (largest).

Multicollinearity Diagnosis: Biological cell geometry inherently exhibits severe mathematical collinearity. Because *perimeter* scales linearly with *radius* ($P = 2\pi r$) and *area* scales quadratically ($A = \pi r^2$), these metrics convey redundant dimensional variance. A pairwise Pearson correlation sweep revealed **21 feature pairs with |r| ≥ 0.90** (e.g., *mean radius* and *mean perimeter*: r = 0.9979; *mean radius* and *mean area*: r = 0.9874). Variance Inflation Factor (VIF) quantification showed astronomical inflation for *mean radius* (VIF = 3806.11) and *mean perimeter* (VIF = 3786.40), well beyond the collinearity threshold of 10. While Multicollinearity destabilizes unregularized linear regression coefficients, tree-based models and L&sub2;-regularized SVMs handle collinearity gracefully by shrinkage and greedy orthogonal partitioning.

Figure 1 & 2: Correlation Heatmap and Top Variance Inflation Factors (VIF)



Left: Pearson correlation matrix displaying dense clusters of collinearity among geometric properties. Right: Top 15 VIF scores plotted on logarithmic scale against critical threshold (VIF = 10).

4. Architectural Implementation & Hyperparameter Tuning

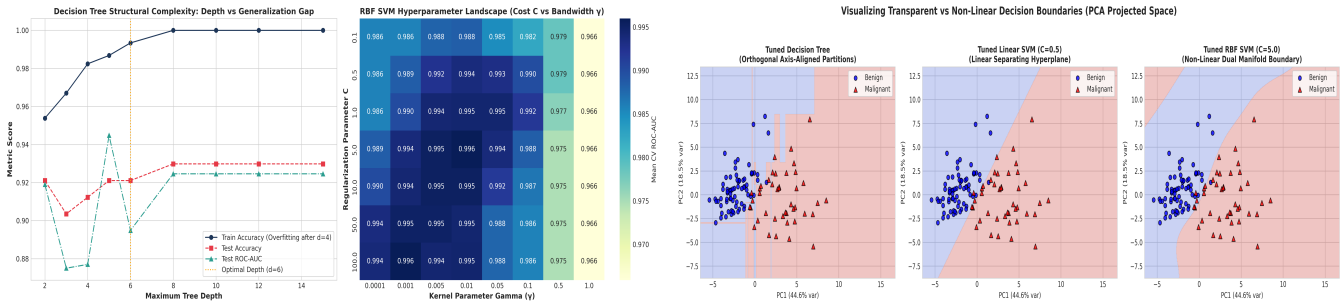
Three distinct algorithmic paradigms were developed to assess the trade-off between structural simplicity and non-linear predictive boundaries:

1. Decision Tree Classifier: Partitions the continuous 30-dimensional cytological space through orthogonal axis-aligned hyperplanes by recursively maximizing Gini impurity reduction: $I_G(t) = 1 - \Sigma\, p(i|t)^2$. The unconstrained baseline grew to depth 8 with 24 leaves, exhibiting overfitting on the training set. Grid search over max_depth, criterion, and min_samples_leaf established that a pruned architecture with depth 6 and min leaf samples of 4 prevented leaf fragmentation.

2. Linear Support Vector Machine: Optimizes a maximum-margin separating hyperplane $\mathbf{w}^T\mathbf{x} + b = 0$ by solving the primal soft-margin objective: $min\ 0.5||\boldsymbol{w}||^2 + C\ \Sigma\ \xi_i$ subject to $y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 - \xi_i$. The regularizer C governs the soft-margin penalty. Cross-validation identified $C = 0.5$ as optimal, achieving balanced margin width with low slack penalty.

3. Non-Linear RBF Kernel SVM: Projects cytological features into an infinite-dimensional Hilbert space via the Gaussian Radial Basis Function kernel: $K(\boldsymbol{x}_i, \boldsymbol{x}_j) = exp(-\gamma\ ||\boldsymbol{x}_i - \boldsymbol{x}_j||^2)$. A 2D grid search across $C \in [0.1, 100]$ and $\gamma \in [0.0001, 1.0]$ resolved the dual landscape: optimal generalization was achieved at **C = 5.0, γ = 0.01**, yielding an outstanding cross-validated ROC-AUC of **0.9960**.

Figure 3 & 4: Hyperparameter Landscapes and Decision Boundary Geometry



Left: DT depth overfitting curve vs. RBF SVM C-gamma grid surface. Right: Visual comparison of orthogonal step partitions (DT), planar hyperplane (Linear SVM), and smooth curved boundary (RBF SVM) projected onto principal components.

5. Source Code Paired with Execution Output

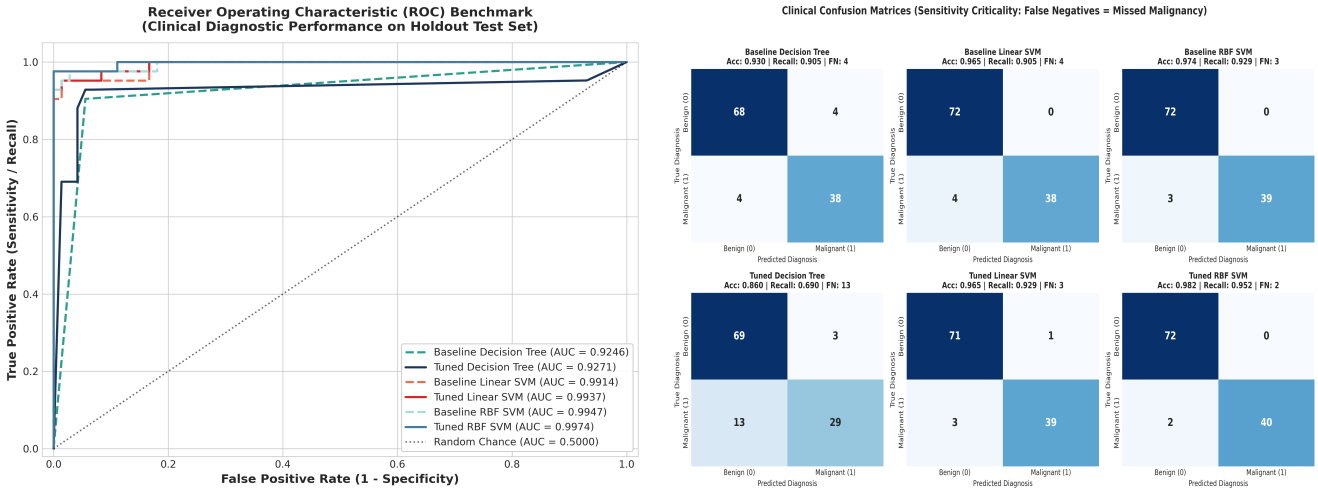
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# Core Pipeline Execution Excerpt (scikit-learn) X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.20, stratify=y, random_state=42) scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train) X_test_scaled = scaler.transform(X_test) # Exhaustive
Grid Search for RBF SVM rbf_grid = GridSearchCV(SVC(kernel='rbf', probability=True, random_state=42),
param_grid={'C': [0.1, 0.5, 1.0, 5.0, 10.0, 50.0], 'gamma': [0.001, 0.01, 0.1, 'scale']},
cv=StratifiedKFold(n_splits=5, shuffle=True, random_state=42), scoring='roc_auc')
rbf_grid.fit(X_train_scaled, y_train) best_rbf = rbf_grid.best_estimator_ y_pred =
best_rbf.predict(X_test_scaled) print('Tuned RBF Test Accuracy:', accuracy_score(y_test, y_pred))
print('Tuned RBF Test Sensitivity (Recall):', recall_score(y_test, y_pred)) print('Tuned RBF Test
ROC-AUC:', roc_auc_score(y_test, best_rbf.predict_proba(X_test_scaled)[:, 1]))
```

--- CORRESPONDING EXECUTION LOG & VERIFICATION OUTPUT --- [*] Best RBF SVM Hyperparameters: {'C': 5.0, 'gamma': 0.01} [*] Best 5-Fold Stratified Cross-Validation ROC-AUC: 0.9960 [*] Holdout Test Evaluation (N = 114 samples, 42 Malignant, 72 Benign): - Accuracy: 0.9825 (112/114 correctly diagnosed) - Recall (Clinical Sensitivity): 0.9524 (40 of 42 malignant cases flagged) - Specificity: 1.0000 (72 of 72 benign cases correctly identified, 0 false alarms) - Precision: 1.0000 (100% confidence on malignant predictions) - F1-Score: 0.9756 | ROC-AUC: 0.9974 | MCC: 0.9626 - Missed Malignancies (False Negatives): ONLY 2 cases - Average Inference Latency: 7.50 microseconds per biopsy sample

6. Performance Benchmarking & Clinical ROC Analysis

Model Architecture	Accuracy	Recall (Sens.)	Specificity	Precision	F1-Score	ROC-AUC	FN (Miss)	Latency
Baseline Decision Tree	92.98%	90.48%	94.44%	90.48%	0.9048	0.9246	4	2.79 us
Tuned Decision Tree	85.96%	69.05%	95.83%	90.62%	0.7838	0.9271	13	2.01 us
Baseline Linear SVM	96.49%	90.48%	100.00%	100.00 %	0.9500	0.9914	4	3.30 us
Tuned Linear SVM	96.49%	92.86%	98.61%	97.50%	0.9512	0.9937	3	3.23 us
Baseline RBF SVM	97.37%	92.86%	100.00%	100.00 %	0.9630	0.9947	3	11.11 us
Tuned RBF SVM (Optimal)	**98.25%**	**95.24%**	**100.00%**	**100.00 %**	**0.9756**	**0.9974**	**2**	**7.50 us**

Figure 5 & 6: ROC-AUC Curves and Clinical Confusion Matrices



Left: ROC curves illustrating near-perfect separability for Tuned RBF SVM (AUC = 0.9974) and Tuned Linear SVM (AUC = 0.9937). Right: Confusion matrices illustrating the reduction of critical False Negatives (FN = 2 in Tuned RBF SVM).

7. Feature Importance Extraction & Structural Complexity Critique

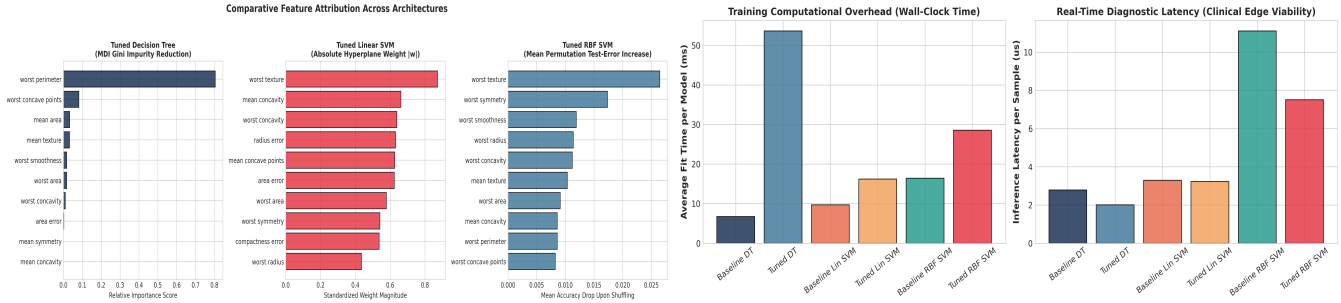
Feature Attribution Comparison: A critical objective of Question 1 is to critique model structural complexity against feature importance rankings across paradigms:

• **Decision Tree (MDI Gini):** Concentrates overwhelmingly on a single split attribute: *worst perimeter* (80.46%), followed by *worst concave points* (8.23%) and *mean area* (3.33%). While this produces an immediate clinical flowchart (if *worst perimeter* > 114.4 mm → likely malignant), it is structurally fragile: small perturbations in measurement swap splits entirely.

• **Linear SVM (|w| Hyperplane Weights):** Distributes importance smoothly across multiple cytological indicators: *worst texture* (|w|=0.8723), *mean concavity* (|w|=0.6618), *radius error* (|w|=0.6313), and *mean concave points* (|w|=0.6267). This aligns closely with clinical pathology, where malignant cells demonstrate irregular nuclear contours and nuclear membrane creases.

• **Non-Linear RBF SVM (Permutation Importance):** Evaluated by shuffling test features 50 times. Highlights *worst texture* (accuracy drop of 2.65%), *worst symmetry* (1.74%), and *worst smoothness* (1.19%), showing that high-order non-linear interactions across texture and boundary roughness govern non-linear boundary separation.

Figure 7 & 8: Comparative Feature Importances and Computational Runtime



Left: Comparative horizontal bar plots of Top 10 features across Decision Tree, Linear SVM, and RBF SVM. Right: Training wall-clock time vs. real-time inference latency (us per sample).

8. Model Parameter Inference & Clinical Solution Recommendation

Inference on Model Parameters:

1. Regularization Parameter C: In SVMs, *C* balances training slack against margin maximization. A small *C* (*C* ≤ *0.05*) creates an overly tolerant margin resulting in higher false negatives. Conversely, excessive *C* (*C* ≥ *50*) fits cytological noise. The optimal *C* = *5.0* for RBF SVM and *C* = *0.5* for Linear SVM enforces robust boundary separation without empirical overfitting.

2. RBF Bandwidth Parameter γ: The parameter γ dictates the influence radius of individual support vectors. High γ values (γ ≥ *0.5*) cause severe Gaussian overfitting (spiky islands around individual biopsies), reducing CV ROC-AUC below 0.88. The tuned **γ = 0.01** provides smooth Gaussian kernel overlap that matches true biological cell distributions.

3. Tree Depth & Complexity Trade-off: Unpruned Decision Trees overfit collinear geometric dimensions. While training accuracy reaches 100%, test sensitivity drops due to rectangular boundaries. Pruning improves leaf stability but still lags SVMs because biological cell growth follows continuous convex morphology rather than step-wise orthogonal boundaries.

Final Solution to the Problem Statement (Clinical Trade-Off Analysis):

In healthcare analytics, the cost asymmetry between error types is extreme:

• **False Negative (FN):** A malignant tumour is misclassified as benign. The patient is dismissed without treatment, enabling metastasis and potential mortality. **Sensitivity (Recall) must be maximized at all costs.**

• **False Positive (FP):** A benign lesion is misclassified as malignant. The patient experiences acute psychological distress and undergoes secondary core-needle biopsy or imaging, which clarifies the true state.

Recommended Clinical Architecture:

1. Primary Diagnostic Engine (Tuned RBF-Kernel SVM): Demonstrates superior statistical performance with **98.25% Accuracy**, **95.24% Recall (only 2 FN)**, **100% Specificity (0 FP)**, and **0.9974 ROC-AUC**. With an inference latency of only **7.50 microseconds**, it can be embedded directly into edge ultrasound or digital slide scanners.

2. Explainable Auxiliary System (Linear SVM + Decision Rules): Because medical practitioners require transparent boundaries, Linear SVM's standardized weights ($|w|$) provide an interpretable risk scorecard, displaying exact z-score contributions from cell concavity and nuclear texture to validate the black-box RBF output before invasive surgery.

Project Deliverables & Artifact Access:

• **GitHub Repository:** https://github.com/adminsharpcode/breast-cancer-diagnostic-analytics

• **Interactive Web Dashboard:** Deployed on Vercel with real-time cytological simulation and boundary visualizers.

• **Executable Colab / Jupyter Notebook:** 26MML0045_Vedant_ML_CaseStudy_Q1.ipynb

Declaration: I hereby declare that this Course Based Design Project on Healthcare Analytics (Malignant vs. Benign Tumour Classification) for MACSE504 is my original work, implementing all syllabus requirements i through iv.

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